

Electronic Dice

1 Problem Description

The goal is to implement an electronic dice with seven LEDs and one push button, simulating a random dice roll. Many systems of this type are battery powered, so battery lifetime is a critical factor for whether the product is usable. The problem is to analyze how using different LEDs, with at least three different colors, affects the total power consumption of the system. The power consumption is examined for the six possible outcomes, during the rolling phase while the button is held down, and as the expected average power consumption during use. The power consumption of the FPGA chip itself is kept outside the analysis.

The overall problem is shown in Figure 1. The figure shows the dice as a system where both the button press and the choice of LED colors are included before the power consumption is analyzed.

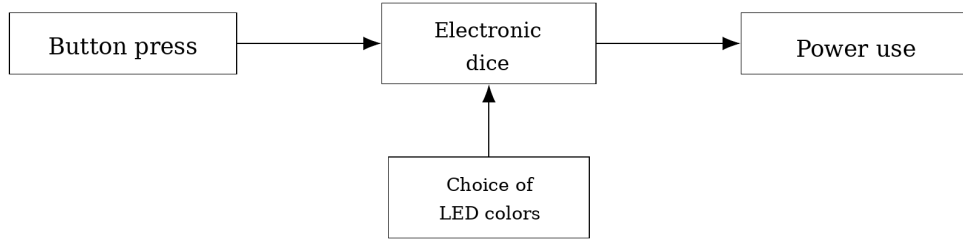


Figure 1: Overall problem figure for the electronic dice and power consumption.

2 Principle

2.1 Overall System Architecture

To simulate randomness, the system is divided into two main logical blocks: a number generator and a decoder, as shown in Figure 2. The number generator is a cyclic counter that, controlled by a clock, continuously counts from 1 to 6 as long as the button is held. Given a sufficiently high clock frequency, the user cannot predict which value the counter will have when the button is released. When the button is released, the counter is frozen at the current value. This 3-bit value is sent to the decoder, which drives the corresponding LEDs so that they show a dice roll.

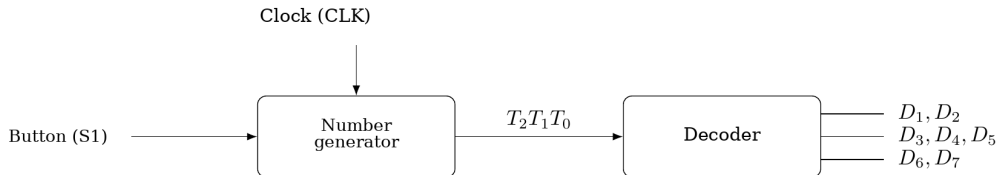


Figure 2: Principle block diagram of the digital system.

2.2 Logic and Truth Table

The relation between the counter value, represented as a 3-bit binary number with bits T_2 , T_1 , and T_0 , and the LEDs that should light to represent the dice is given in Table 1.

Table 1: Overview of LED states depending on the dice roll.

Roll	T_2	T_1	T_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
1	0	0	1	OFF	OFF	OFF	ON	OFF	OFF	OFF
2	0	1	0	ON	OFF	OFF	OFF	OFF	OFF	ON
3	0	1	1	ON	OFF	OFF	ON	OFF	OFF	ON
4	1	0	0	ON	ON	OFF	OFF	OFF	ON	ON
5	1	0	1	ON	ON	OFF	ON	OFF	ON	ON
6	1	1	0	ON	ON	ON	OFF	ON	ON	ON

2.3 Analog Equivalent Circuit and Power Formulas

For the theoretical calculation, the FPGA output pins ($V_{\text{out},1}$ to $V_{\text{out},7}$) are modeled as ideal voltage sources without internal resistance ($R_{\text{int}} = 0 \Omega$). Figure 3 shows the general circuit diagram for the outputs, independent of specific voltage values and component colors.

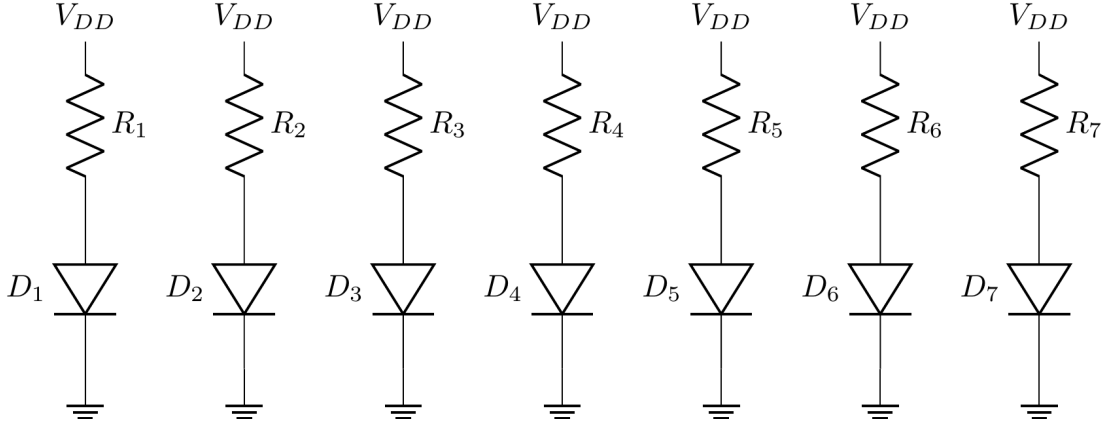


Figure 3: General analog equivalent circuit for the LEDs. The output pins are modeled as ideal voltage sources that provide V_{DD} at a logical high value.

The current $I_{D,i}$ through an active branch is determined by the supply voltage V_{DD} and the diode forward voltage $V_{f,i}$, using the resistor relation:

$$I_{D,i} = \frac{V_{DD} - V_{f,i}}{R_i} \quad (1)$$

The total power P_{tot} consumed by the LEDs and series resistors for a specific dice roll is the sum of the power in the branches that are turned on:

$$P_{\text{tot}} = \sum_{i \in \text{active}} V_{DD} \cdot I_{D,i} \quad (2)$$

If a fair dice is assumed, where all six outcomes are equally likely with probability $p = 1/6$, the expected average power consumption $E[P]$ can be calculated as the expectation value:

$$E[P] = \frac{1}{6} \sum_{k=1}^6 P_k \quad (3)$$

Here, P_k is the total power consumption when the dice shows value k .

3 Implementation and Test

3.1 Circuit Schematic for Physical Wiring

For the physical implementation, a Go Board with a Lattice iCE40 FPGA was used. To examine how LED color affects power consumption, two red LEDs (D_1, D_2), three yellow/orange LEDs (D_3, D_4, D_5), and two green LEDs (D_6, D_7) were used. Figure 4 shows how all components and colors were connected. A $10\text{ k}\Omega$ pull-down resistor was placed between the push button S_1 and ground. This makes sure that the FPGA input pin is pulled to a defined logical low level when the button is open and prevents a floating input. All seven LEDs used identical $570\text{ }\Omega$ series resistors.

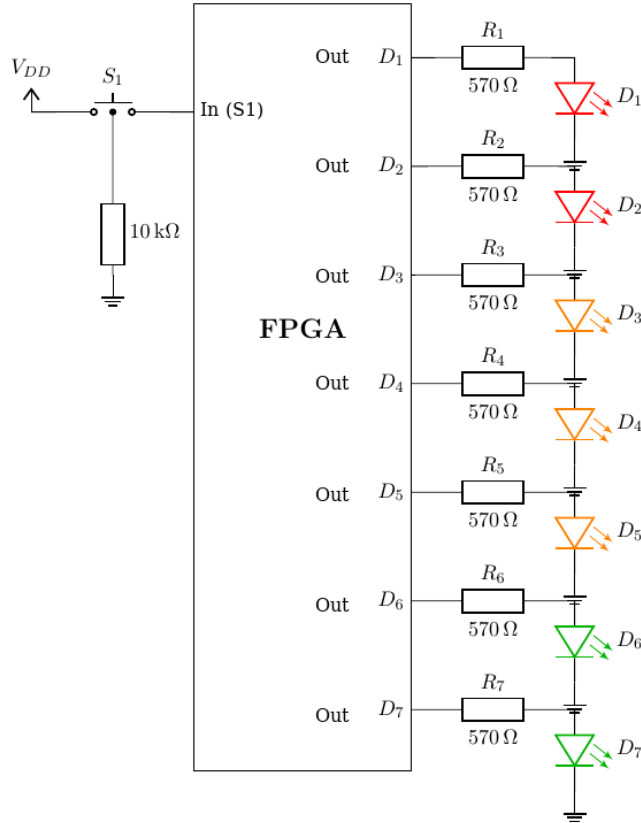


Figure 4: Physical circuit schematic for the dice wiring, showing all seven LEDs. The supply voltage from the FPGA is $V_{DD} = 3.3\text{ V}$.

The completed wired solution is shown in Figure 5.

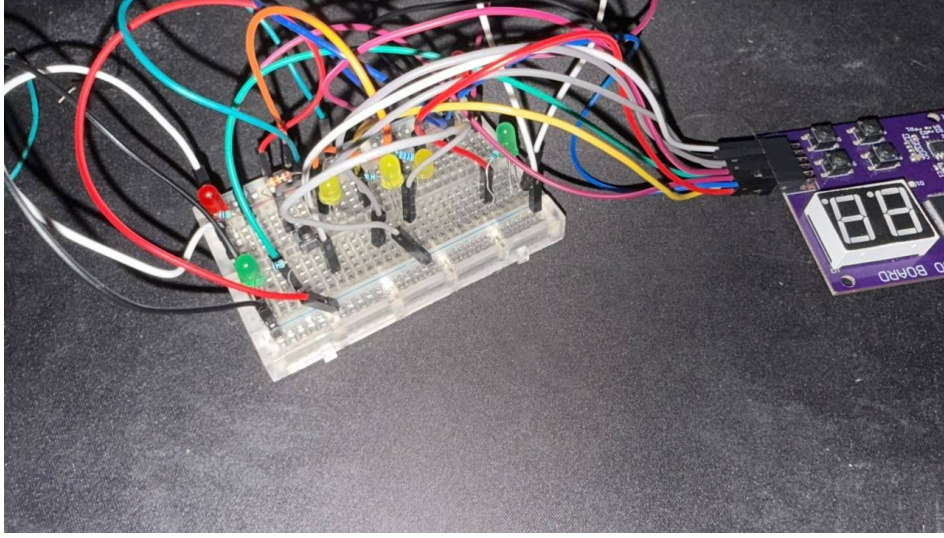


Figure 5: Photograph of the completed system on the breadboard.

3.2 FPGA Programming

The digital logic for the system was modeled and programmed in Icestudio. The clock generator uses the integrated hardware clock on the FPGA board, 25 MHz. This clock was connected through a prescaler module where the division parameter was set to $N = 12$. This results in the cyclic counter updating at a frequency of

$$\frac{25 \text{ MHz}}{2^{12}} \approx 6.1 \text{ kHz}.$$

A speed of approximately six thousand rotations per second makes the outcome appear random to the user. The finished logic implementation in Icestudio is shown in Figure 6.

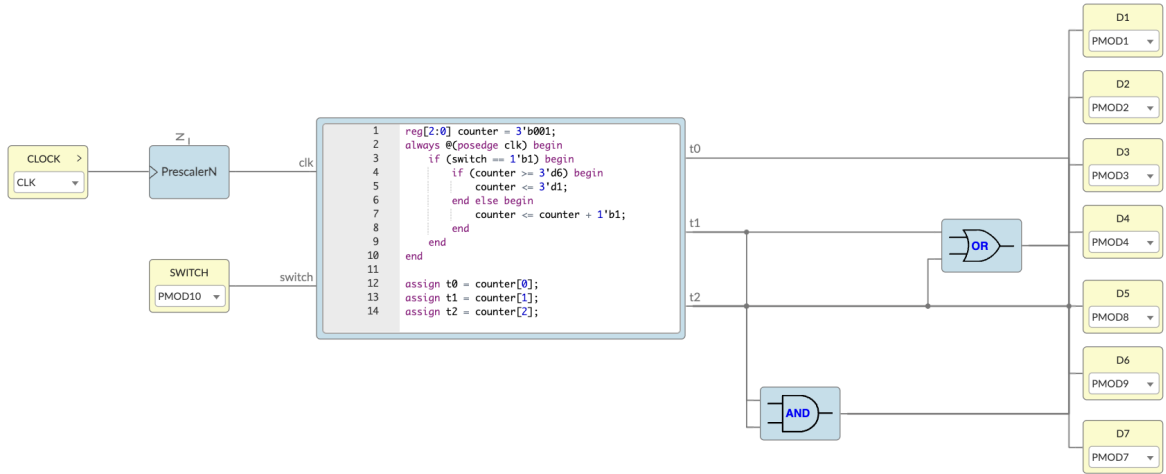


Figure 6: Implementation of hardware clock, prescaler, number generator, and decoder logic in Icestudio.

3.3 Theoretical Calculations and LTspice Simulation

Before the physical measurements were performed, the circuit was simulated in LTspice to estimate the theoretical power consumption with actual component values. Since the three LED colors have different semiconductor properties, they also have different forward voltages V_f , which leads to different current draw in the branches. With $V_{DD} = 3.3 \text{ V}$ and series resistors of $R = 570 \Omega$, the simulation showed that the red LEDs drew approximately 3.08 mA, the yellow LEDs 2.84 mA, and the green LEDs 2.36 mA. The simulation setup is shown in Figure 7, and the calculated values are summarized in Table 2.

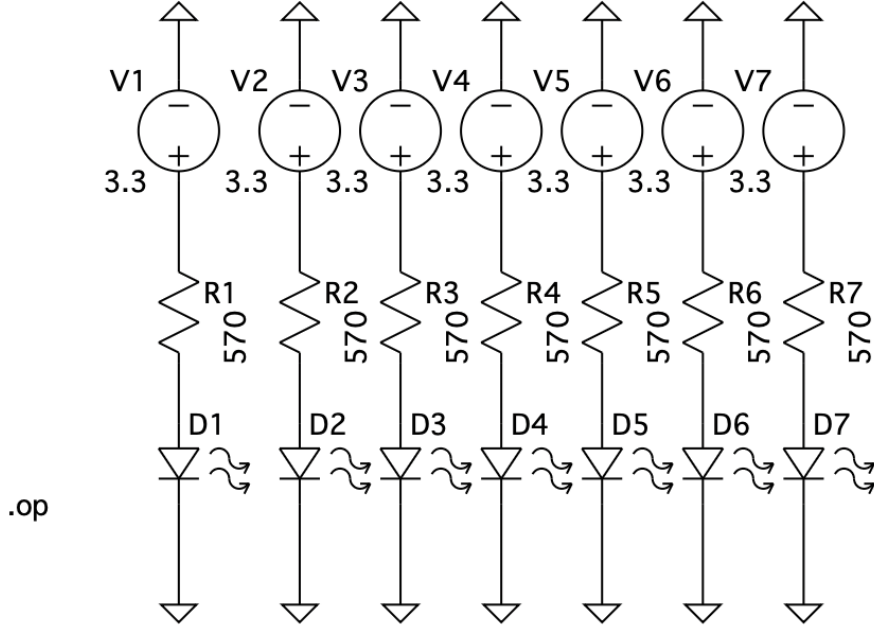


Figure 7: Specific model of the LEDs and series resistors in LTspice with correct values for V_{DD} (3.3 V) and resistance (570 Ω).

Table 2: LTspice-simulated current and power consumption at 3.3 V.

Outcome	Active LEDs	Total current (mA)	Total power (mW)
1	D_4	2.84	9.37
2	D_1, D_7	5.44	17.95
3	D_1, D_4, D_7	8.28	27.32
4	D_1, D_2, D_6, D_7	10.88	35.90
5	D_1, D_2, D_4, D_6, D_7	13.72	45.28
6	All except D_4	16.56	54.65

3.4 Physical Measurements With Multimeter

To validate the theoretical model, the total current through the active LED branches was measured physically with a multimeter for each dice roll. The results are presented in Table 3. Deviations between simulation and physical measurements are mainly caused by the FPGA I/O pins not acting as perfect ideal sources in practice. They have some internal resistance, which means that the voltage drops slightly under load. In addition, the series resistors have tolerances of $\pm 5\%$.

Table 3: Physically measured values with multimeter.

Outcome	Active LEDs	Measured total current (mA)	Calculated power (mW)
1	D_4	2.67	8.81
2	D_1, D_7	5.19	17.13
3	D_1, D_4, D_7	7.86	25.94
4	D_1, D_2, D_6, D_7	10.38	34.25
5	D_1, D_2, D_4, D_6, D_7	13.05	43.07
6	All except D_4	15.72	51.88

Expected power consumption: Assuming the dice is fair, all values occur equally often. Based on the physical measurements in Table 3, the expected average power consumption while a result is displayed is

$$E[P] = \frac{181.08}{6} = 30.18 \text{ mW} \quad (4)$$

Power consumption while the button is held down: When the user presses the switch S_1 , the counter rotates very quickly with the clock frequency of 6.1 kHz. During this phase, the LEDs draw the expectation value calculated above on average, 30.18 mW. In addition, the current path through the button pull-down resistor is closed, where the 3.3 V supply is applied across 10 k Ω . This gives an additional constant power consumption:

$$P_{\text{pull-down}} = \frac{(3.3 \text{ V})^2}{10 \text{ k}\Omega} = 1.09 \text{ mW} \quad (5)$$

Based on the physically measured LED-branch powers in Table 3 and the calculated contribution from the pull-down resistor, the power consumption while the button is held down, with the FPGA chip kept outside the calculation, is estimated as

$$P_{\text{button down}} = 30.18 \text{ mW} + 1.09 \text{ mW} = 31.27 \text{ mW} \quad (6)$$

4 Conclusion

The electronic dice was implemented and tested so that power consumption from the LEDs could be quantified. LTspice gave a theoretical maximum consumption of 54.65 mW for outcome six, while the corresponding physical measurement gave 51.88 mW. The expected average power consumption while a result is displayed was calculated as 30.18 mW, and during the roll itself this increased to 31.27 mW because of current through the pull-down resistor.

The deviation between simulation and measurement is small and can be explained by non-ideal I/O pins and tolerances in the series resistors. The results therefore show how LED color affects current consumption and give a practical basis for evaluating required battery capacity in a similar implementation.